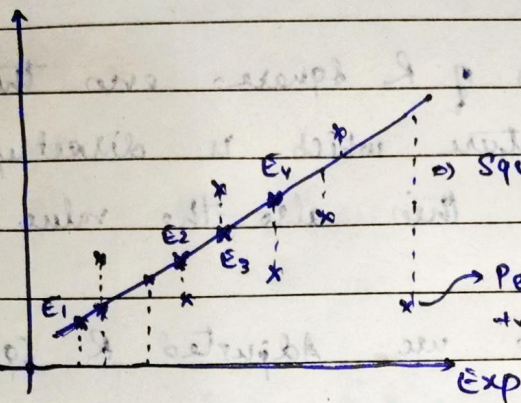


★ MSE, MAE, RMSE [Cost function] → Performance metrics

1. MSE

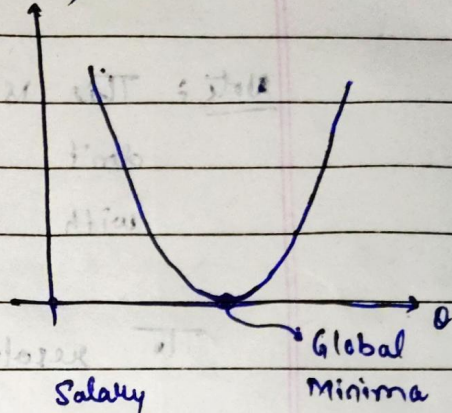
Salary



⇒ Squaring

Penalties the outlier

$J(\theta_0, \theta_1)$

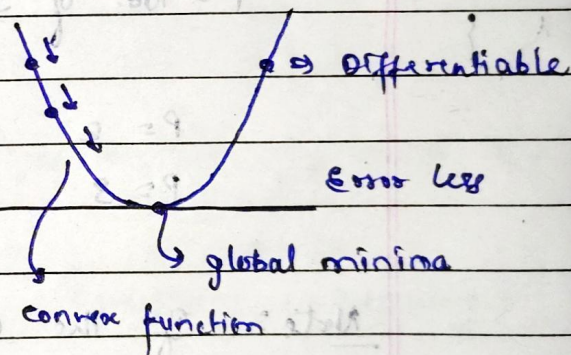


$$MSE = \sum_{i=1}^n \frac{(y - \hat{y})^2}{n} \rightarrow \text{cost function}$$

Quadratic Equation

$$ax + by + c$$

$$(a-b)^2 = a^2 - 2ab + b^2$$



Error less

global minima

convex function

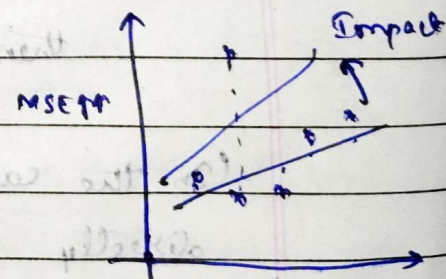
Slope 50

converge

local minima

global minima

Non-convex functions



Advantage

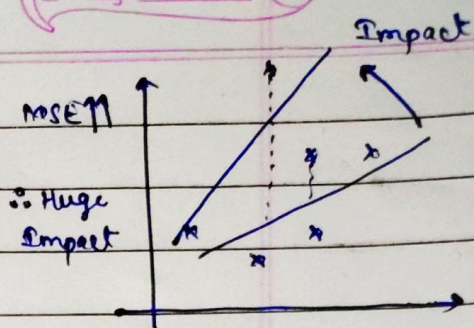
Disadvantage

- I. Differentiable
- II. It has one local and one global minima
- III. Converge faster

- I. Not Robust to outliers
- II. It is not in same unit.

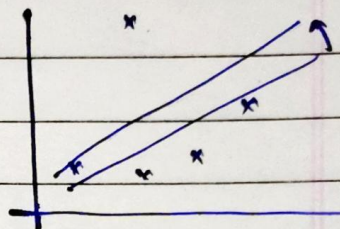
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$\frac{1}{n}$ (loss)
 $(y_i - \hat{y}_i)^2$ (loss)²



ii. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

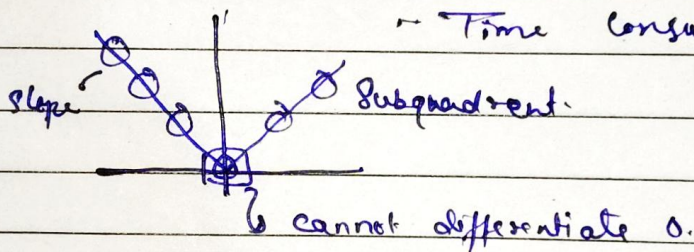


Advantage

- i. Robust to Outliers
- ii. It will be in the same unit

Disadvantage

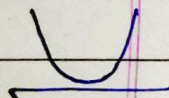
- i. Convergence usually take more time. Optimization is a complex task.
- ii. Time consuming.



iii. RMSE (Root Mean Squared Error).

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



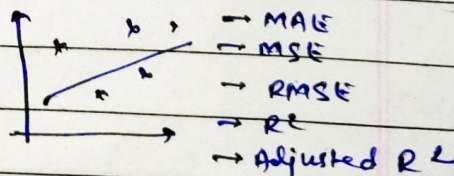
Adv.

- i. Same Unit
- ii. Differentiable

Disadv.

- i. Not Robust to Outliers.

Note: At starting we focus all of these $\{MSE, MAE, RMSE\}$



Interview:-

MSE vs MAE vs RMSE
 R^2 vs Adjusted R^2